

20. Answer: C

Input power to the inverter

$$P = IV$$

$$= 5.7 \times 280$$

$$= 1596 \text{ W}$$

Since power loss is 10%, output power of inverter

$$P_{\text{out}} = 0.9 \times 1596$$

$$= 1436 \text{ W}$$

$$I_{\text{r.m.s.}} = \frac{P_{\text{out}}}{V_{\text{out}}}$$

$$= \frac{1436}{230}$$

$$= 6.2 \text{ A}$$